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Proterozoic rotation of the Peter Lake domain in the Trans-Hudson orogen of Saskatchewan, Canada, deduced from paleomagnetism

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Abstract

The Peter Lake domain is a northeast-trending, 200 km long, cigar-shaped terrane. On its southeast side is the 1854 ± 11 Ma Wathaman batholith that marks the suture zone for the juvenile terranes of the Trans-Hudson orogen (THO), and on its northwest side is the Wollaston fold belt that defines the margin of the Archean Slave–Rae–Hearne (SRH) craton. Two well-defined remanent magnetization components are carried by hematite, magnetite and pyrrhotite in a range of metamorphic rock types. The A component was probably acquired during the D1 regional metamorphic event at ~1869 Ma and gives a direction of $D=302°$, $I=-40°$ ($\alpha_{95}=10°$, $k=16$, $N=15$). Its pole position at 3°N, 52°W ($d_p=7°$, $d_m=12°$) locates the Peter Lake domain along the margin of the SRH craton at the time of metamorphism and was later rotated 32° ± 14° counterclockwise, probably when the abutting southwestern arm of the Wathaman batholith was similarly rotated during oroclinal deformation. The B component is found preferentially in the Parker Lake gneisses but is also found in minor shear zones in the Peter Lake complex and Wathaman batholith, and it replaces the A component by remagnetization. B has a direction of $D=92°$, $I=83°$ ($\alpha_{95}=8°$. $k=19$, $N=20$) that gives a pole at 54°N, 80°W ($d_p=15°$, $d_m=15°$). The B remanence is thought to have been acquired at about 1780 Ma during the D3 shearing event. This event produced the Needle Falls and Parker Lake shear zones that separate the Peter Lake domain from the Wollaston fold belt and Wathaman batholith, respectively. The B component likely defines the end of the Trans-Hudson orogenic event during which the 5000 km wide Manikewan ocean closed. Thorough evaluation of the Andean collisional model for the THO will require much more paleomagnetic data.

1. Introduction

Several geologic syntheses over the past decade have concluded that the Trans-Hudson orogen (THO) in Saskatchewan and Manitoba was formed in an Andean-type convergent collision of the Archean Slave–Rae–Hearne (SRH) and Superior cratons (e.g. Gibb et al., 1980; Green et al., 1985; Hoffman, 1988). This resulted in the closure of the intervening Manikewan ocean to entrap the intervening Proterozoic juvenile terranes, to generate the Wathaman batholith as the ensialic magmatic arc at the suture zone, and to form the bounding fold belts in the cratonic margins (Fig. 1; Gibb, 1983; Stauffer, 1984; Thomas and Gibb, 1985; Lewry et al., 1990; Hoffman, 1990). Using paleomagnetic data for the 1825 Ma to 1900 Ma time period, Dunsmore and Symons (1990) noted that the minimum width of the Manikewan ocean basin was at least 4800 km (>43°). Paleomagnetic study of the Wathaman batholith has shown further that it was formed at

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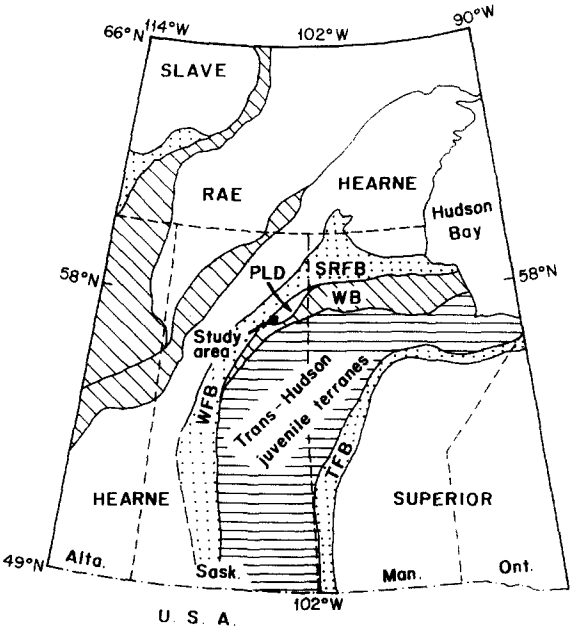


Fig. 1. Location of the study area with the main tectonic elements from Hoffman (1989) including *PLD* (Peter Lake domain), *SRFB* (Seal River fold belt), *TFB* (Thomson fold belt), *WB* (Wathaman batholithic arc) and *WFB* (Wollaston fold belt).

the margin of the SRH craton, supporting the suture interpretation, and that the outboard La Ronge–Lynn Lake domain, Flin Flon domain and Superior province were translated relatively northwestward by about 1200 km (11° ± 11°), 3000 km (27° ± 12°) and 5500 km (49° ± 16°) into the suture, respectively, as the ocean closed (Symons, 1991). The Peter Lake domain is a "suspect" Archean terrane located between the Wathaman batholith and SRH craton but separated from them by major faults. This study uses paleomagnetism to study the tectonic evolution of the domain and, thereby, to further test and refine the Andean-type closure model for the THO. Noting that only three paleomagnetic studies have been published to date on rock units from the THO, much paleomagnetic work remains to be done to track domain motions in time and space during closure.

2. Geology

The Peter Lake domain is a northeasterly-trending cigar-shaped terrane about 200 km in length and 50 km in width. It is separated on its northwestern side from the Wollaston fold belt of the SRH craton by the Needle Falls shear zone, and on its southeastern side from the Wathaman batholith by the Parker Lake shear zone. Both zones are *en échelon* ductile mylonitic shears with dextral strike-slip displacements (Lewry and Collerson, 1990). The domain is comprised of medium- to coarse-grained plutonic rocks of dioritic to gabbroic composition with minor metapyroxenite and anorthosite that have been intruded by granitic rocks ranging from monzonite to alaskite in composition (Lewry et al., 1980). In the study area (Fig. 2), the mafic rocks are mostly weakly foliated gabbros, metagabbros and amphibolites, and the granitic rocks are quartz-bearing monzonites to biotite monzogranite. These units were metamorphosed in the regional D1 deformational event. D2 was a minor localized event that preceded emplacement of the Wathaman batholith. Adjacent to the Parker Lake shear zone, these units have been tectonized to strongly foliated biotite hornblende gneiss and biotite granite gneiss. The shear zone itself contains intensely-banded flasered augen gneisses, submylonites and blastomylonites. Lewry et al. (1980) attributed most of the shearing in the domain to the D3 event that postdates emplacement of the unmetamorphosed massive Wathaman batholith. Excluding zones of dynamic metamorphism, the regional metamorphic grade is in the upper greenschist to lower amphibolite facies. Retrograde metamorphism in the shear zones belongs to the lower greenschist facies and the associated minor movement is termed the D4 event. The D5 event is the last event and is related to brittle NNE-faulting. D4 and D5 may be in part coeval.

Ray and Wanless (1980) first noted the "suspect Archean" age of the Peter Lake domain when the Parker Lake gneisses from 10 km west of site 22 gave a K–Ar biotite age of 1689 ± 10 Ma, a K–Ar hornblende age of 2177 ± 110 Ma, and a U–Pb zircon age of 2538 ± 10 Ma (see

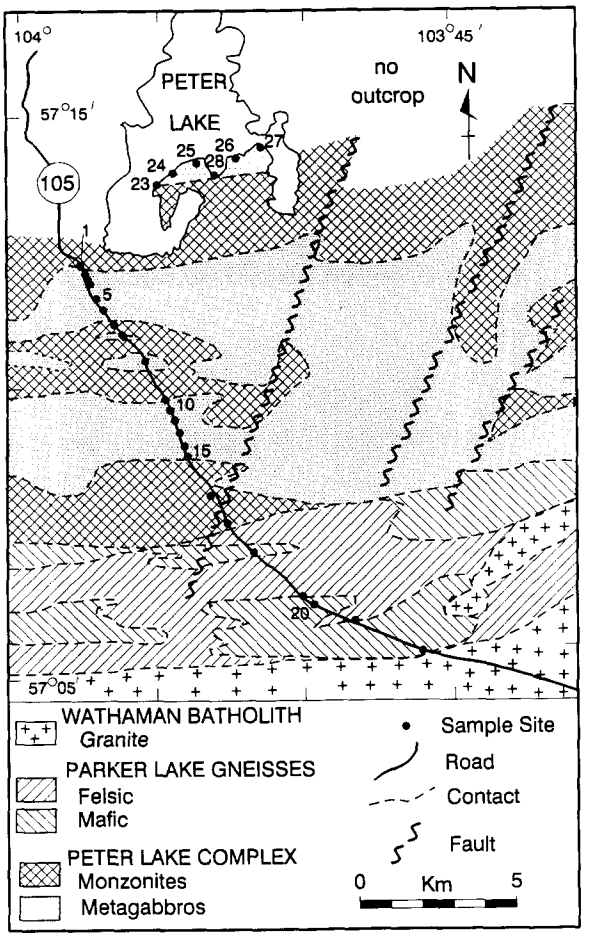


Fig. 2. Geology of the study area simplified after Lewry et al. (1980), with site locations.

Slimmon et al., 1991, for a compilation of Saskatchewan age dates). Bickford et al. (1986, 1987, 1988) have reported U–Pb zircon ages for granites of 1840±29 Ma and 2780±160 Ma, for granitic and tonalitic gneisses of 1860±5 Ma, 2186±46 Ma, 2321±95 Ma, 2390±160 Ma, 2556±22 Ma and 2582±19 Ma, and for gabbroic pegmatite of 1908±27 Ma. Only the 2780 Ma age comes from the study area and is on Peter Lake. Of the other eight age dates, four come from the south end of Wollaston Lake and four from the north end of Reindeer Lake about 60 km and 125 km to the northeast, respectively. This spectrum shows clearly that the Peter Lake domain is Late Archean in origin, that it contains younger Proterozoic plutons, and that these rocks have been subsequently variably metamorphosed until as late as about 1.7 Ga.

3. Experimental methods

The Peter Lake domain was sampled at 28 sites along the Highway 105 transect and the shoreline of Peter Lake (Fig. 2). Of these sites, twenty are in the metagabbros and three in the monzonites of the Peter Lake complex. The remaining five sites are in the Parker Lake mafic gneisses to provide paleomagnetic contact tests between the Wathaman batholith and the Peter Lake domain, and between the gneisses and the complex in the domain.

Five cores were drilled at most sites and oriented in situ by sun compass. Two specimens were later sliced from most cores to yield a total collection of 259 specimens. All measurements were done inside a shielded room. The specimens' natural remanent magnetizations (NRM) were measured using an automated CTF DRM-420 cryogenic magnetometer. One specimen from each site was alternating field (AF) demagnetized in twelve steps to 120 mT using a Sapphire Instruments SI-4 demagnetizer, and a second specimen was thermally demagnetized in twelve steps to 680°C using a Schonstedt TSD-1 demagnetizer. The remaining specimens were AF demagnetized in four or more steps, mostly at 15 to 20 mT increments above 15 or 20 mT, followed by one thermal step at 625°C for residual hematite remanence for about half of the specimens to define the characteristic remanent magnetization (ChRM) of each specimen (Table 1). Accepted ChRM directions have maximum angular deviations of less than 10° and mostly of less than 3° (Kirschvink, 1980), except for a few specimens for which remagnetization circle directions were also used (Bailey and Halls, 1984). The ChRM directions were averaged to get the mean directions of the magnetization component(s) for each site (Fisher, 1953; Table 1). Saturation isothermal remanent magnetization (SIRM) tests were run on 10 specimens by pulse-magnetizing them with a Sapphire Instruments

Table 1
Site mean remanence statistics

Site no.	Rock type	Number spec.			Demagnetization range used		Mean direction				
		e	r	o	mT	°C	Decl. (°)	Incl. (°)	α_{95} (°)	k	M
1	gb	8	0	0	20–100	500–600	88	86	9	35	B
2	gb	4	2	2	20–60	450–600	298	−21	19	11	A
		2	0	0	20–60		11	77		21	B
3	gb	8	0	1	20–120	450–670	318	−32	11	28	A
4	gb	5	0	0	15–80	500–670	272	−43	27	9	A
		2	0	0	20–60		225	14		41	C
5	gb	1	1	0	15–45		309	−29		7	A
		2	0	2	15–45	400–560	353	75		47	B
6	gb	6	0	0	15–80	500–580	299	−39	12	30	A
7	mo	6	0	0	20–60	400–500	22	85	11	36	B
8	gb	10	0	0	15–45	500–560	24	81	5	112	B
9	gb	7	0	0	20–120	500–670	261	9	7	77	C
10	gb	6	0	2	15–60	200–300	358	82	19	13	B
13	gb	2	1	7	20–50	200–300	309	−42	8	48	A
14	gb	7	4	0	20–80	200–300	315	−38	7	40	A
15	gb	7	1	3	15–120	0–400	313	−45	12	21	A
16	mb	10	0	0	20–120	200–600	308	−39	9	30	A
17	mo	5	2	3	20–80		322	−25	11	32	A
18	mg	4	0	0	20–80	300–530	237	−53	8	121	A
		6	0	0	20–80		144	56	7	84	B
19	mg	7	0	0	20–120	500–580	201	0	16	16	C
		2	0	0	40–80		106	73		141	B
20	mg	8	1	1	20–120	0–400	314	−35	8	41	A
21	mg	7	0	3	20–80	300–500	113	63	9	21	B
22	mg	5	1	4	20–120	200–300	34	73	14	19	B
23	gb	8	0	0	20–100	400–600	101	58	11	17	B
		1	0	0	20–100		279	−43		11	A
24	gb	9	0	1	20–120	500–600	137	69	12	19	B
25	gb	8	0	0	20–100		353	77	11	26	B
		2	0	0	20–80	530–600	301	−19		24	A
26	gb	5	0	2	20–60		215	32	12	40	C
		2	0	0	40–80	500–580	308	−70		60	A
27	gb	9	0	1	20–80	500–600	240	51	15	13	C
28	gb	9	0	0	20–80	450–670	253	55	12	21	C
		2	0	0	15–80		154	71		48	B

Note. Rock types are gabbro (gb), monzonite (mo), mafic blocks or bands in monzonite (mb) and mafic gneiss (mg). Number of specimens giving used endpoints (e), used remagnetization circles (r), and not used (o). The mean direction is given by its declination (Decl.), inclination (Incl.), radius of cone of 95% confidence (α_{95}) and precision parameter (k) (Fisher, 1953). The M column lists the defined magnetization component. Sites 11 (gb) and 12 (mb) gave no coherent data.

SI-6 unit in eleven steps to 900 mT, then AF-demagnetizing them in six steps to 50 mT.

4. Results

The ChRM site mean directions form three groups, two with geologic control and the third without evident control. The A ChRM component is found most commonly in gabbroic sites of the Peter Lake complex and is directed to the west-northwest at intermediate-up inclinations (Fig. 3a). The B component is found mostly in the Parker Lake gneisses and other sheared rocks, and is directed to the east at steep-down inclina-

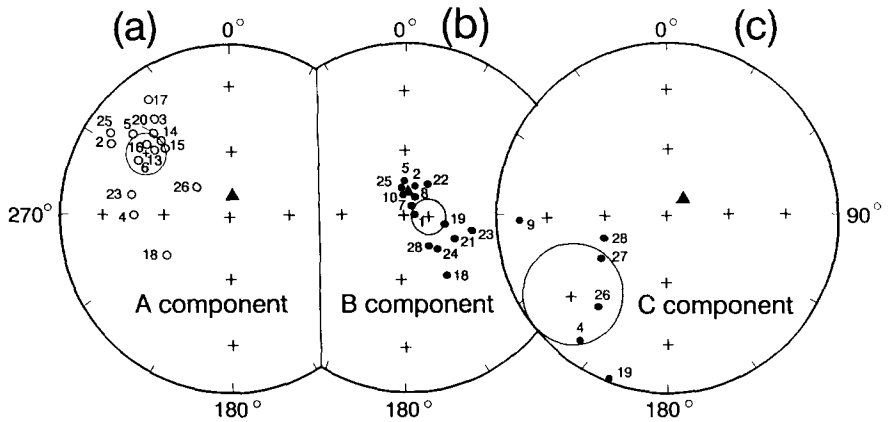


Fig. 3. Equal-area stereogram showing the site mean directions (circles) and unit mean directions (cross with cone of 95% confidence) for the A, B and C magnetization components. The mean for B includes six sites in the Wathaman batholith that are not shown. Solid (open) symbols are plotted in the lower (upper) hemisphere. Triangle is the present Earth's magnetic field direction.

tions (Fig. 3b). The C component is simply a label to describe several site mean ChRM directions scattered in the southwest down quadrant (Fig. 3c). Two sites (#11 and #12) gave no coherent directions.

4.1. A component

The A component is the sole or dominant remanence preserved in nine of seventeen gabbroic sites and two of three monzonitic sites from the Peter Lake complex. Conversely the A component is found in only one of five sites from the mafic Parker Lake gneisses. This distribution suggests that A is a metamorphic remanence in the complex. The A component is routinely well isolated by AF step demagnetization in the coercivity spectrum over 20 mT (Figs. 4a, 4b). Thermal step demagnetization shows that A is retained by domains that are preferentially unblocked in the 300–400°C, 500–600°C or 600–700°C temperature ranges that are characteristic of pyrrhotite, magnetite and hematite, respectively (Figs. 4c–4e). Many of these A specimens combine two of these three magnetic minerals with pyrrhotite and hematite being particularly prevalent. SIRM testing shows that the A component resides in domains ranging from nearly single-domain (SD) to multidomain (MD) sizes (Fig. 5, specimens 3, 6, 15, 16). Unfortunately the SIRM characteristics of pyrrhotite, which is prevalent in these rocks, are not adequately known; characteristic curves are not available. Two of the four A component curves (Fig. 5, specimens 15 and 16) are probably "SD" pyrrhotite and two (Fig. 5, specimens 3 and 6) are probably MD magnetite, based on thermal demagnetization data for the site. The basic observations, however, for the A component are that it is found in the Peter Lake complex: (1) along the full length of the HW 105 transect; (2) in the several rock types found along the transect; and (3) in all three common magnetic minerals with unblocking temperatures from 300°C to 700°C and with SD to MD size characteristics. This combination of features is typical of a remanence component acquired during regional metamorphism to upper greenschist–amphibolite grade.

The A component site mean directions were divided into two groups for statistical comparison; those represented by four or more specimens give statistically the same mean direction as those represented by two or three specimens at the 95% confidence level. Therefore all A site mean directions were combined into one overall mean direction (Table 2).

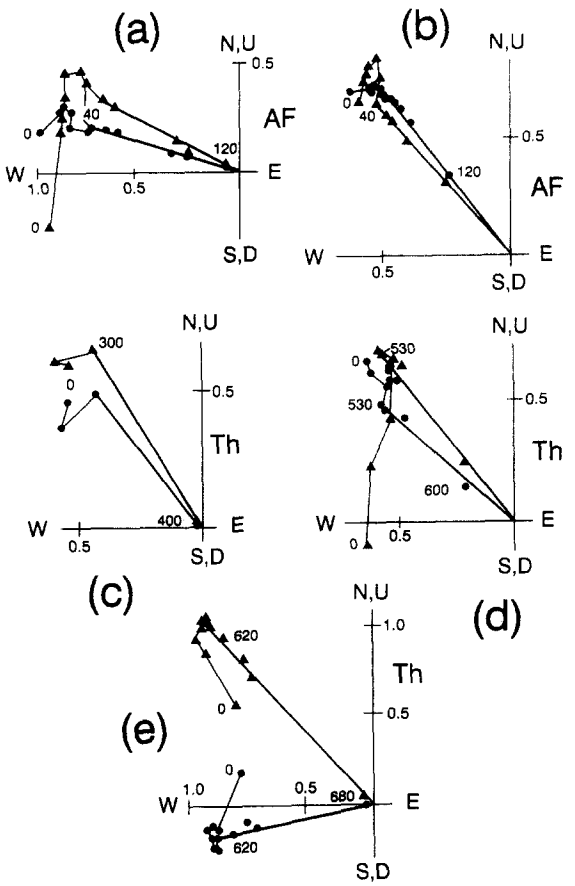


Fig. 4. Orthogonal plots of the decay of the remanence vector on alternating field (*AF*) and thermal (*Th*) step demagnetization for example A component specimens from sites: (a) 6; (b) 16; (c) 15; (d) 6; (e) 4. Points on the horizontal (*N, E, S, W*) and vertical (*U, D*) planes are denoted by circles and triangles, respectively. The axial intensities are expressed as a ratio of the NRM intensity (J/J_0) with the initial J_0 values being: (a) 88.0, (b) 0.554, (c) 0.174, (d) 188, and (e) 77.7×10^{-5} A/m, respectively. *AF* step field intensities are in mT and thermal step temperatures (*Th*) in °C.

4.2. B component

The B ChRM is the sole or dominant component in seven of seventeen gabbroic sites and one of three monzonitic sites from the Peter Lake complex. More significantly it is the main component in three of five sites in the Parker Lake mafic gneisses.

The B component is easily defined by AF demagnetization although the contribution of viscous remanent magnetization (VRM) from the present Earth's magnetic field (PEMF) to B in a given specimen is not easily estimated as they are about parallel in direction (Figs. 3b, 6a, 6b). Thermal step demagnetization shows that the B ChRM resides in pyrrhotite, magnetite and hematite (Figs. 6c–6e), usually in a combination of two of these magnetic minerals. SIRM testing shows that B also resides in minerals with SD to MD characteristics (Fig. 5, numbers 6, 8, 22, 24). Like the A component, B is found along the length of the HW 105 transect in the Peter Lake complex, is represented in the several rock types, and is found in the three main magnetic minerals. The major difference from A which is predominant in the gabbros of the Peter Lake complex, is that the B component is predominant in the Parker Lake mafic gneisses. This suggests that B is a metamorphic component associated with the D3 shearing event, and this interpretation is supported by the identification of B in shear zones of the Wathaman batholith as discussed later.

As done for the A component, the B site mean directions represented by four or more specimens were compared with those with less than four specimens and found to be statistically the same at the 95% confidence level. Therefore all of the B site mean directions were combined to determine the B component unit mean direction (Table 2).

4.3. C component

Four gabbroic sites and one mafic gneiss site retain the C remanence component as the sole or dominant component. Its relatively rare occurrence suggests that C is not as important as the A or B component in the geologic evolution of the Peter Lake domain. The C component is easily isolated by AF demagnetization once an initial VRM from the PEMF is removed (Figs. 7a, 7b). Thermal step demagnetization shows that C is carried mostly by magnetite and hematite (Figs. 7c, 7d). SIRM analysis shows the presence of pseudosingle-domain (PSD) to MD magnetic minerals (Fig. 5, numbers 9 and 27). Of the four gabbroic sites carrying the C component, three

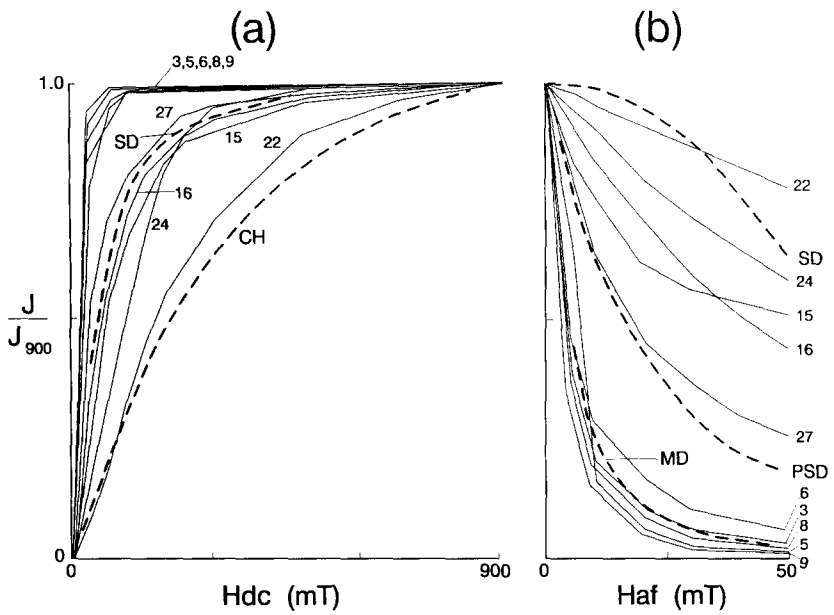


Fig. 5. Saturation isothermal remanent magnetization (SIRM) test specimen results showing (a) acquisition of SIRM, and (b) AF demagnetization of SIRM relative to type curves for single-domain (*SD*), pseudosingle-domain (*PSD*) and multidomain (*MD*) magnetite and for coarse-grained hematite (*CH*). J/J_{900} is the ratio of the measured remanence intensity to SIRM intensity at 900 mT, and H_{dc} and H_{af} are the applied magnetizing and demagnetizing fields, respectively. The numbers refer to the specimen's site.

Table 2
Unit mean remanence statistics and poles

Magnetization component/sites		Mean direction					Pole position			
		N	Decl. (°)	Incl. (°)	α_{95} (°)	*k*	°N	°W	d_p (°)	d_m (°)
A	2–4, 6, 13–18, 20	11	303.3	−39.5	11.4	17.0				
A	5, 23, 25–26	4	298.2	−40.7	28.7	11.2				
A	above sites	15	302.1	−39.8	9.8	16.2	3.3	52.3	7.1	11.8
B	1, 7, 8, 10, 18, 21–25	10	95.1	79.1	10.7	21.3				
B	2, 5, 19, 28	4	76.5	83.4	19.7	22.7				
B	above sites	14	91.5	80.4	8.5	23.0	52.3	72.2	15.6	16.3
B	above sites + Wathaman sites	20	91.8	83.1	7.6	19.4	54.2	80.0	14.6	14.9
C	4, 9, 19, 26–28	6	232.1	28.9	28.5	6.5	−5.6	153.7	17.3	31.3

Notes. *N* is number of sites. "Mean direction" defined in Table 1. The "Pole position" is given by its latitude, longitude, and radii of 95% confidence (d_p, d_m) in degrees (Fisher, 1953).

(26, 27, 28) are from a restricted area at the southeast corner of Peter Lake (Fig. 2). All three have west-southwesterly declinations with intermediate-down inclinations that differ from the shallow inclinations of the other C component sites. The obvious speculation is that this restricted area has undergone substantial tectonic rotation or a later localized remagnetization event.

4.4. Relative age of A and B components

Most A component specimens show the initial removal of a steeply-inclined downward magnet-

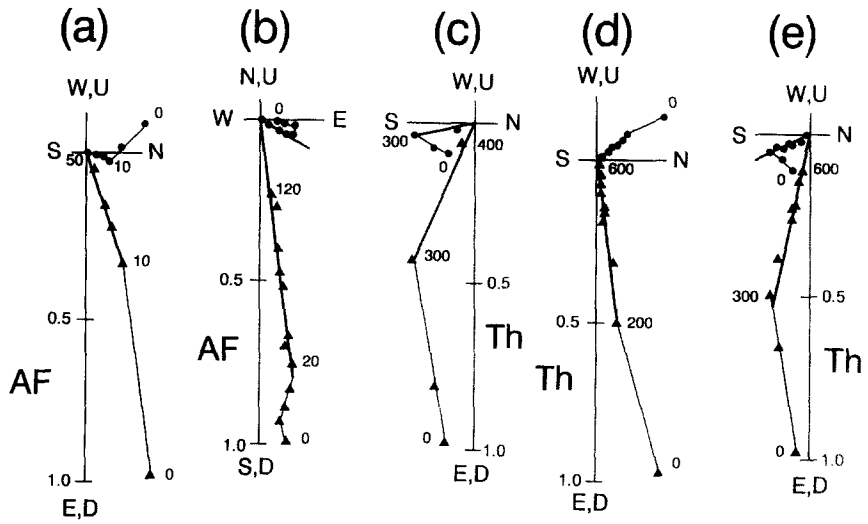


Fig. 6. Orthogonal plots of example specimens having the B magnetization component from sites: (a) 5; (b) 24; (c) 22; (d) 8; (e) 1. J_0 is: (a) 124, (b) 7.85, (c) 56.9, (d) 131, (e) 0.0995×10^{-5} A/m, respectively. Conventions as in Fig. 3.

ization component on step demagnetization (Figs. 4a, 4b, 4d, 4e). It could be the initial removal of a B component in lower-coercivity or unblocking-temperature domains, or it could simply be the removal of VRM from the PEMF, or it could be a combination of both the B component and VRM. There is no evidence of the prior removal of an A component from B ChRM specimens.

More interesting are the ChRM directions from sites 17 to 22. They define a crude great circle path going from the A to B component direction (Fig. 8), corresponding to progressively more intense levels of shearing. Lewry et al. (1980) note that "in general, mylonitic zones become larger and more common towards the contact with the Wathaman batholith" in the Parker Lake gneisses. Similarly the specimens themselves show a variable but increasing level of foliation and lamination. This implies that the A component is older and is being replaced by the B component.

5. Discussion

5.1. The A component pole

The pole position for the A ChRM mean direction of the Peter Lake domain is located at 3°N, 52°W ($d_p = 7°$, $d_m = 12°$) (Table 2, PLD-A in Fig. 9a). It is close to the pole of the SRH craton for the 1825–1900 Ma interval and to the 1854 Ma Wathaman batholith pole. The A component is found in pyrrhotite, magnetite and hematite that reflect unblocking temperatures from low to high metamorphic grades, and it is found in the several disparate rock units in the Peter Lake complex. It is sensible, therefore, to correlate acquisition of the A magnetization component with the one significant metamorphic event that has affected these rocks, the D1 regional metamorphic event. The later D2 event was minor and very localized. The D1 event predates intrusion of the 1854±11 Ma Wathaman batholith which is not foliated by it. The 1854±11 Ma age is the average and root mean square error derived from six U/Pb zircon dates (Ray and Wanless, 1980; Meyer, 1987; Van Schmus et al.,

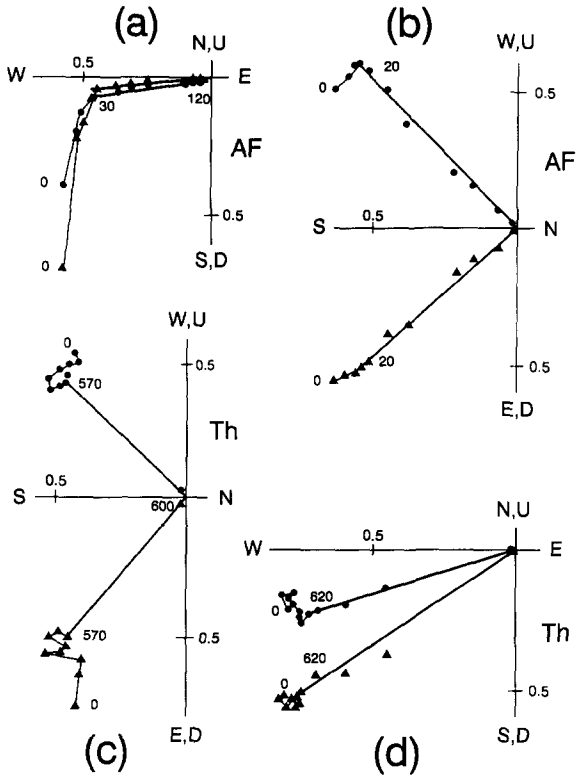


Fig. 7. Orthogonal plots of example specimens having the C magnetization component from sites: (a) 9; (b) 26; (c) 27; (d) 28. J_0 is: (a) 174, (b) 2.69, (c) 1.40, (d) 3.91×10^{-5} A/m, respectively. Conventions as in Fig. 3.

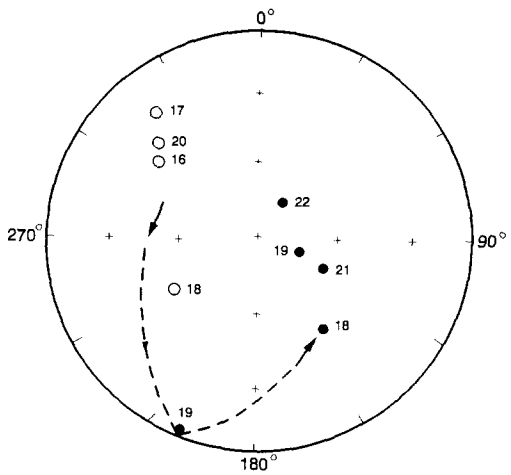


Fig. 8. Site mean ChRM directions along HW 105 from site 16 in the Peter Lake complex across the Parker Lake gneisses to site 22 next to their contact with the Wathaman batholith.

1987). Further it appears that the D1 event occurred between about 1855 Ma and 1900 Ma, correlating with the youngest cluster of U–Pb zircon age dates of 1840±29 Ma, 1860±5 Ma and 1908±27 Ma from the domain. The average of these three ages, 1869±29 Ma, is taken as the best estimate of the acquisition of the A remanent magnetization component and, therefore, of the D1 regional metamorphic event. No younger K–Ar, Ar–Ar or Rb–Sr dates are yet available for the Peter Lake complex.

The Peter Lake A pole locates the domain at a paleolatitude of 22°±6° when the A magnetization was acquired, and this value is statistically the same as the 28°±9° paleolatitude for the SRH craton during the 1825–1900 Ma interval (Beck, 1989; Fig. 9a). This is the prime evidence to conclude that the Peter Lakes domain is a rotated fragment, with possible minor translation, of the SRH craton and not a relict of the Superior or some other Archean craton. The A pole also indicates that the Peter Lake domain was rotated counterclockwise by 32°±14° about a vertical axis relative to the SRH craton.

The 32°±14° counterclockwise rotation of the Peter Lake domain is similar in sense and only somewhat larger than the about 15° counterclockwise rotation found for the Wathaman batholith relative to the SRH craton (Symons, 1991). Symons (1991) suggested that the 15° rotation of the Wathaman batholith could be caused by oroclinal bending of the southwestern arm of the boomerang-shaped batholith relative to its northeastern arm (Fig. 1). A rotation of about 23° counterclockwise would be consistent with the displacement of both the Peter Lake A and Wathaman A poles relative to the SRH pole within their error limits. As a further speculation, the rotation could have been caused by impact of the THO terranes into the SRH craton. Relative counterclockwise rotation of the Peter Lake domain by displacements on the boundary Needle Falls and Parker Lake shear zones is unlikely because kinematic indicators point to dextral strike-slip motion on both zones (Lewry and Collerson, 1990) which would cause clockwise rotation. Contrary to statements made in Sy-

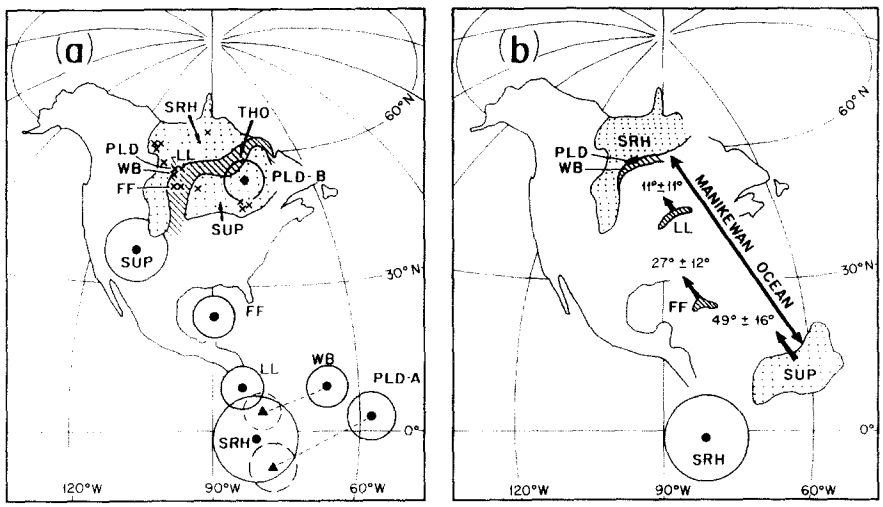


Fig. 9. Pole positions with their cones of circular standard deviation for 1825–1900 Ma rock units (×) in the Superior (*SUP*) and Slave–Rae–Hearne (*SRH*) provinces and Trans-Hudson orogen (*THO*) (*PLD-A*=Peter Lake domain (A component); *FF*=Flin Flon domain; *LL*=Lynn Lake-La Ronge domain; *WB*=Wathaman batholith) in: (a) present position, and (b) restored to about 1865 Ma position with the arrows and angles indicating the amount of motion involved in the restoration with its error. The paleomagnetic and age dating information for these poles, excluding PLD from this study, are listed in Table 3 and are discussed fully in Dunsmore and Symons (1990) and summarized in Symons (1991). The triangles show the PLD and WB poles (A components) after correction in paleodeclination for 31° and 15° counterclockwise rotation, respectively, about a vertical axis. The location of the pole for the B component in the Parker Lake gneisses of the Peter Lake domain is shown by a diamond.

Table 3
Pole positions for 1825–1900 Ma rock units in and adjacent to the Trans-Hudson orogen

Pole	Units	Long. (°W)	Lat. (°N)	CSD (°)	Ref.
SRH	Slave, Rae, Hearne provinces	82	−2	8	1
PLD-A	Peter Lake domain, A component	52	3	7[b]	2
PLD-B	Peter Lake domain, B component	80	54	11[b]	2
WB	Wathaman batholith	67	9	3[b]	3
LL	Lynn Lake–La Ronge domain	84	9	4[b]	4
SUP	Superior province	110	38	8	5

Notes. 1=Average of the 1835 Ma Martin Formation (Evans and Bingham, 1973), 1825 Ma Dubawnt Group (Park et al., 1973), 1850 Ma Kahochella Group (Reid et al., 1981), 1870 Ma Seton Formation (Irving and McGlynn, 1979), and 1890 Ma Akaitcho Formation (Evans et al., 1980) poles. 2=This study. 3=1854 Ma Wathaman batholith pole (Symons, 1991). 4=Average of 1855 Ma Amisk metavolcanics and Boundary intrusion poles (Park, 1975). 5=Average of the 1838 Ma Spanish River complex (Robertson and Watkinson, 1974), 1849 Ma Sudbury granophyre (Morris, 1984), 1850 Ma Sudbury norite (Morris, 1984) and 1884 Ma Molson dikes (Ermanovics and Fahrig, 1975) poles. CSD is circular standard deviation based on pole ([b] site) mean directions.

mons (1991), sinistral rather than dextral motion on the shear zones is required to explain the discordance of the Peter Lake and Wathaman A poles from the SRH pole by translation. Tilt about a horizontal axis is not a likely explanation for the counterclockwise rotation because the axis would have to be about perpendicular to the northeast trend of the Trans-Hudson domains in Saskatchewan.

5.2. The B component pole

The mean direction for the B remanence component yields a pole located at 52°N, 72°W ($d_p=16°$, $d_m=16°$). This B component is best developed in the Parker Lake gneisses, the intensely sheared equivalent of the Peter Lake rock units (Lewry et al., 1980). The intense shearing was the result of the D3 event that postdates emplacement of the Wathaman batholith. The Parker Lake B component is also statistically identical to the Wathaman batholith's C component at 64°N, 105°W ($d_p=33°$, $d_m=34°$) which is found in sheared granite sites (Symons, 1991), especially in the four sites from immediately south of the batholith's contact with the gneisses on the HW 105 transect. Combining the data from the fourteen B component sites of this study with the six equivalent C sites from the Wathaman study gives an overall pole at 54°N, 80°W ($d_p=15°$, $d_m=15°$). This pole falls on the 1650 Ma position of the Irving (1979) apparent polar wander path (APWP). This age is certainly too young because the D3 event must have preceded the onset of deposition of the undeformed Athabaska sandstones at about 1735±15 Ma that conformably overly the Wathaman batholith about 35 km west of site 22 (Lewry et al., 1980; Hoffman, 1989; Machado, 1990). It is possible that the B component in the shear zones was affected by retrograde metamorphism in the later D4 or final D5 deformation events. Some evidence that the B magnetization component is more complex and, perhaps, represents more than one event is seen in Fig. 3b. The mean directions from sites 1, 2, 5, 7, 8, 10, 22 and 25 could be forming one cluster with a different direction ($D=13°$, $I=80°$, $\alpha_{95}=4°$, $k=163$, $N=8$) and pole (79°N, 88°W, $d_p=8°$, $d_m=8°$) from the cluster formed by the means from sites 18, 19, 21, 23, 24 and 28 ($D=125°$, $I=66°$, $\alpha_{95}=10°$, $k=50$, $N=6$; 25°N, 67°W, $d_p=13°$, $d_m=16°$). Further rock properties work being done on these rocks as part of the Lithoprobe project may shed some light on the one versus two component problem. For the present, they are simply grouped as one B population. Note further that the D4 and D5 events must predate 1735±15 Ma also, but they are minor compared to the D3 event and unlikely to leave a regional ChRM overprint. Given the shear zone environment, it is probable that the B component in hematite, magnetite, and pyrrhotite is a chemical remanent magnetization. The K–Ar biotite and hornblende ages of 1669±40 Ma and 2177±110 Ma from the Parker Lake gneisses (Ray and Wanless, 1980) are not sufficiently concordant to provide an indication of the B component's age. Of interest, however, are the results of Annesley and Madore (1992), from about 14 km north of site 25 in the Peter Lake domain next to the Needle Falls shear zone. U–Pb zircon ages on protolith give 2076±3 Ma and 2068 Ma with a reset titanite age of 1810 Ma. A Rb–Sr whole rock age of 1765±30 Ma is reported by Bell and Macdonald (1982) for the Middle Lake granites. These are late granitoid plutons that occur in the northeastern half of the Peter Lake domain. Another indication comes from Cumming and Scott (1976), who report four Rb–Sr whole rock ages that average 1751±66 Ma from metamorphic rocks in the Wollaston fold belt across the Needle Falls shear zone from the Peter Lake domain. Money (1968; Slimmon et al., 1991) has reported six K–Ar biotite, muscovite and hornblende ages from schists and gneisses in the Wollaston fold belt which, excluding one divergent value, average 1784±57 Ma. Finally Annesly and Madore (1991) have reported several ages from sheared rocks in the Wollaston fold belt immediately across the Needle Falls shear zone from the Peter Lake domain including U/Pb zircon lower intercepts of 1783±27 Ma and $^{207}Pb/^{206}Pb$ sphene ages of 1778, 1779 and 1789 Ma that indicate a relatively strong 1780–1790 Ma event. Some of these ages may reflect regional uplift, but some probably reflect the shearing event. Therefore from these various values and limits, 1780±30 Ma is thought to provide the best estimate for the acquisition age of the B metamorphic component.

5.3. C directions

Noting the very scattered nature of the C site mean ChRM directions and their restricted geo-

graphic distribution outside the small peninsula on Peter Lake containing sites 26 to 28, little significance should be attached to the C pole position. It is possible that the C remanence is an earlier primary or secondary metamorphic remanence that was not overprinted by the A or B metamorphic ChRM components. When the C pole is calculated from the average of all six C site mean directions, it is located at 6°S, 154°W or at the antipodal 6°N, 26°E. Neither pole appears to correlate with any of the few pre-1900 Ma paleomagnetic poles from the SRH craton (Irving et al., 1984; Dunsmore and Symons, 1990), and the same conclusion is reached when the pole is calculated for sites 26 to 28 only. It may also be that C is simply various proportions of the A plus B components, reflecting various degrees of remagnetization. The C ChRM site means do fall approximately on the great circle shown in Fig. 8. If this option is true then the C mean direction and its pole are meaningless.

6. Summary

The metamorphosed gabbro-monzonite rock suite of the Peter Lake complex retains a common A ChRM direction in magnetic domains with unblocking temperatures ranging from about 300 to 700°C. The A component predates intrusion of the adjacent massive 1854±11 Ma Wathaman batholith, and correlates most evidently with the D1 regional metamorphic event in the Peter Lake domain. The best estimate for the age of the D1 event and of the A component is 1869±29 Ma based on scant radiometric data, but it is unlikely to be older than the onset of the Hudsonian orogeny at about 1900 Ma. The A pole agrees with the coeval SRH pole in paleoinclination showing that the Peter Lake domain was beside the SRH craton at about 1870 Ma, but disagrees in paleodeclination showing that the domain was rotated 32°±14° counterclockwise relative to the craton after about 1870 Ma. Dextral strike-slip kinematic indicators on the domain-bounding Needle Falls and Parker Lake shear zones argue strongly against either some 3000 km of left-lateral transcurrent fault motion or simple domain rotation about a vertical axis to explain the discordance of the poles. At present the most likely explanation is that the Peter Lake domain underwent oroclinal bending along with the southwest arm of the Wathaman batholith which has a similarly discordant pole, probably as part of the D3 event and possibly reflecting indentation tectonics as the juvenile terranes of the THO were compressed in the final closure of the Superior craton against the SRH craton (Fig. 9b).

The B remanence component is found preferentially in the Parker Lake mafic and felsic gneisses of the Peter Lake domain. It is also found over a broad spectrum of unblocking temperatures, and is evidently a metamorphic component. The B component clearly postdates 1854 Ma as it is found in sheared quartz monzonite of the Wathaman batholith. Acquisition of the B component must predate the 1735±15 Ma deposition of the overlying Athabaska sandstones. Several lines of radiometric age dating information indicate that the best estimate for the age of the B component and for the D3 shearing event is 1780±30 Ma. Unfortunately, because the B component is nearly vertical, it is insensitive to terrane rotation about a vertical axis, and because there are no reliable 1780 Ma poles from the SRH craton, it is not possible to show that the B component postdates the counterclockwise rotation event. When comparable poles are available from the SRH and Superior Archean cratons and from the other domains of the THO, they should coincide to show that these terranes were united into a single craton by about 1780 Ma if the geotectonic model for closure of an Atlantic-sized oceanic basin is correct (Dunsmore and Symons, 1990; Symons, 1991; Fig. 9b).

Overall this paleomagnetic study indicates that the Peter Lake domain existed prior to the onset of the Hudsonian orogeny as part of the SRH craton, was metamorphosed and rotated during the orogeny, and gives results that are in agreement with an Andean-type closure model for the THO. This model is contrary to the assertion of Piper (1991) that the Precambrian Shield was a coherent entity so that one should not expect to

find such plate motions. The overall pattern of poles for the study area as shown in Fig. 9a is obviously difficult to explain in terms of one plate with a convoluted APW path, and so are specific pole comparisons. For example, the 1854 Ma Wathaman pole in the THO (Symons, 1991) is about 45° from the 1850 Ma Sudbury irruptive poles in the Superior craton (Morris, 1984).

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